

Question:1

Draw a line in a diagram from position (1, 3) to (2, 10) then to (6, 12) and finally to position (18, 20). (Mark each point with a beautiful green colour and set line colour to red and line style dotted).

Question:2

PROGRAM 7 : Sarah bought a new car in 2001 for \$24000. The dollar value of her car changed each year as shown in the table below.

Value of Sarah's Car	
Year	Value
2001	\$24,000
2002	\$22,500
2003	\$19,700
2004	\$17,500
2005	\$14,500
2006	\$10,000
2007	\$ 5,800

represent the following information using a line graph with following style properties

- X-axis – year
- Y-axis – car value
- Title – value depreciation (left aligned)
- Line style dash dot & line color should be red
- Point using * symbol with green color & size 20

Question:3

PROGRAM 8: Create scatter plot for the below data (use scatter function)

Product	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Affordable Segment	173	153	195	147	120	144	148	109	174	130	172	131
Luxury Segment	189	189	105	112	173	109	151	197	174	145	177	161
Super Luxury Segment	185	185	126	134	196	153	112	133	200	145	167	110

Create scatter plot for each segment with following properties within one graph

- X-axis – months of year with font size 18
- Y-axis – sales of segments
- Title – sales data
- Color for affordable segment- pink
- Color for luxury segment – yellow
- Color for super luxury segment - blue

Question:4

Write Python programs using Pandas and Matplotlib to perform the following data visualization tasks using the Iris dataset (iris.csv).

- a) Load the iris.csv dataset using Pandas and display the first five records of the dataset.
- b) Create a scatter plot showing the relationship between sepal.length and sepal.width.
- c) Create a scatter plot of the Iris dataset using different colors for each flower variety (Setosa, Versicolor, and Virginica).
- d) Create a line plot showing the relationship between sepal.length and sepal.width.
- e) Create a histogram showing the distribution of sepal.length values.
- f) Create a bar chart showing the total sepal.length for each Iris flower variety.
- g) Create a box plot to visualize the distribution of sepal.length for different Iris flower varieties.
- h) Create a pie chart showing the percentage distribution of the three Iris flower varieties in the dataset.
- i) Create a scatter plot comparing petal.length and petal.width, using different colors for each Iris variety.
- j) Create multiple subplots displaying the following histograms:

Histogram of sepal.length

Histogram of sepal.width

Histogram of petal.length

Histogram of petal.width

- k) Create a grouped bar chart comparing the average sepal.length and sepal.width for each Iris flower variety.

l) Create a bubble scatter plot using `petal.length` and `petal.width`, where the size of each bubble represents `sepal.length`.